

Online Retail II: Revenue and Customer Value Analysis (SQL + Power BI)

A solo project where I am responsible for data loading, cleaning, SQL analysis, and dashboard design.

Objective:

I analyzed a large online retail transactions dataset to understand revenue trends, customer value, and product performance. The project follows a full workflow from raw CSV to SQL cleaning and analysis, then ends with a dashboard designed for business decisions. Role: data analyst responsible for SQL pipeline and dashboard design.

Tools used for this project:

pgAdmin/PostgreSQL for data modeling and SQL

Power BI for dashboard visuals

Excel for quick checks and CSV export

Questions explored throughout this project:

1. How does total revenue change by month and by quarter, and which periods generate the largest share of revenue.
2. Which customers bring the most revenue, what share of total revenue comes from those customers, and how frequently those customers place orders.
3. Which products generate the most revenue, which products sell often but with low revenue per order, and which products underperform on both volume and revenue.
4. Which countries or regions contribute the highest total revenue, and which countries show the highest average order value.
5. What is the average order value, how does order value differ by country or by high value versus regular customers, and how many orders fall into low, medium, or high value ranges.
6. How much revenue comes from first time customers versus repeat customers, and how do order values differ between those two groups.

7. How much revenue is lost due to returns, which products or countries show higher return volume, and how returns change net revenue.
8. How do customer spend tiers compare, for example low, medium, and high spend groups, in terms of revenue share and order frequency.

Data Attributes:

Source

- Name: Online Retail II
- Origin: UCI Irvine: Machine Learning Repository
- # of rows and columns: 541,910 rows and 6 columns

Main fields and meaning

- InvoiceNo
Unique invoice identifier
- StockCode
Product code
- Description
Product name
- Quantity
Number of units purchased
- InvoiceDate
Date and time of the transaction
- UnitPrice
Price per unit in currency
- CustomerID
Unique customer identifier
- Country
Customer country

Limitations

- No demographics
- Only country level location
- Limited time range
- Returns represented through negative quantities

Folder Directory

Dashboard

- Power BI dashboard file (OnlineRetailProject.pbix)
- Power BI PDF export of the dashboard

Data_clean

- CSV files of all data cleaned with SQL

Data_raw

- Original Excel file from the UCI Machine Learning Repository

Databasescreenshots

- Screenshots of queries that create the core tables and confirm that the full Online Retail II dataset loaded into raw_orders in PostgreSQL

Docs

- Log of the main actions taken during the project

QueryResults

- Screenshots of the data output of queries I used to find specific data for each question

Sql

- SQL scripts used to clean the raw data and answer the eight business questions

Key findings

Revenue trend

- Total revenue over the period is about \$10.64M.
- Quarterly revenue grows from about \$0.82M in 2010 Q4 to about \$3.30M in 2011 Q4, which is roughly 31% of total revenue.
- Revenue steps up almost every quarter, so sales trend upward through 2011 instead of staying flat.

Customer value

- The top 10% of customers generate about 61% of revenue.

- The top 20% of customers generate about 75% of revenue, and the top 30% generate about 83%.
- The bottom 50% of customers contribute only about 8% of revenue, so a small group of high value repeat buyers dominates sales.

Product performance

- The top 20 products generate about \$1.64M, roughly 15% of total revenue.
- These top products include postage, shipping and a small set of core gift items that drive both volume and value.
- A long tail of products has low units and low revenue, which adds catalog complexity with limited impact on total sales.

Country performance

- The United Kingdom accounts for about \$9.00M in revenue and 18,786 orders, roughly 85% of total revenue.
- Among countries with at least 100 orders, Ireland has the highest average order value at about \$984, followed by France, Germany and then the UK.
- Several smaller markets have higher average order value but much lower total revenue, so they act as premium but niche markets.

Order value and ranges

- Average order value is about \$514, while the median order value is about \$299.
- Orders range from large negative adjustments around -\$11K up to an extreme invoice around \$168K, so a few outliers pull the average up.
- Using three bands (low < \$50, mid \$50–\$150, high ≥ \$150), low and mid orders together are about 27% of orders but only around 3% of revenue, while high value orders are about 73% of orders and about 97% of revenue.

New versus repeat customers

- First orders generate about 22% of revenue, with an average order value around \$431.
- Repeat orders generate about 78% of revenue, with an average order value around \$497.
- Revenue depends heavily on existing customers, who also spend more per order than new customers.

Returns and net revenue

- Gross revenue from positive lines is about \$10.64M.
- Returns sum to roughly \$0.90M, so net revenue is about \$9.75M and returns absorb about 8% of gross revenue.

- A small set of products and countries shows higher returns ratios, pointing to specific areas for quality, description or logistics review.

Customer spend tiers

- Customers split evenly into low, mid and high tiers by total revenue (about 1,446–1,447 customers per tier).
- Low tier customers contribute about 4% of revenue and place around 1.3 orders on average.
- Mid tier customers contribute about 12% of revenue and place around 2.6 orders on average.
- High tier customers contribute about 85% of revenue and place around 8.9 orders on average, so the top third of customers drives most revenue and repeat activity.

Dashboard overview

The Power BI report is a one page overview with:

- KPI cards for total revenue, net revenue, returns rate, and average order value at the top of the page.
- A column chart for revenue by quarter to show growth across the period.
- A bar chart for revenue by country to compare the United Kingdom with other markets.
- A bar chart for products with the highest returns rate to highlight risk by product.
- A line chart for revenue by month to show the month to month pattern inside the overall trend.
- A bar chart for top products by revenue to show which products drive sales.
- A column chart for average order value by country to compare spend per order across markets.

Skills Demonstrated

- Data cleaning and feature engineering in SQL
- Date handling, window functions and aggregation in PostgreSQL
- Building a small dimensional model around orders, customers and products
- Translating business questions into SQL queries
- Designing a compact dashboard from exported result sets
- Writing project documentation for a technical and recruiter audience

Business impact and next steps

Business impact

The analysis shows heavy revenue concentration in a small group of customers and products, strong dependence on the UK market, and meaningful revenue loss from returns. This points to three priorities. Protect and grow high value customers, improve the performance or positioning of key products, and address high return items or markets.

Next steps

If I extend this project I will:

- Build customer cohorts over time to track retention and changes in spend.
- Estimate customer lifetime value using order history and simple churn signals.
- Add product bundling or basket analysis to understand which items sell together.
- Layer in marketing or campaign data if available to link revenue and retention to specific actions.